FIN ST1812–ST1901

Direct Gauge-Covariant Dirichlet Action,

Topology, Refinement and Spectral-Triple Obstructions

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Six adaptive research rounds — 90 programmes
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This report follows the Release 11.03 falsification of the APD–moment– L_{total} route. It attempts a different construction directly from the strict Dirichlet form: a first-order weighted differential and a locally gauge-covariant lattice matter action. The construction is then immediately tested for uniqueness, curvature, refinement, continuum, chirality, fermions and gravity.

English mathematical and critical research report.

Abstract

ST1812–ST1901 investigate whether the strict FIN Dirichlet form can produce a gauge-covariant action without the previously falsified APD and three-moment routes. The answer is conditionally positive for finite matter dynamics and negative for unique physical gauge geometry.

Because every strict off-diagonal weight is positive, the current support is the complete graph K_{12}. Choosing one orientation per edge defines a weighted incidence operator d_W with

$$A = d_W^* d_W.$$

Introducing $U(1)$ link phases gives the covariant difference

$$(d_U\psi)_{ij} = \sqrt{w_{ij}}(\psi_i - U_{ij}\psi_j)$$

and the action

$$S_{\text{matter}}[\psi, U] = \frac{1}{2}\sum_{i<j} w_{ij}|\psi_i - U_{ij}\psi_j|^2 + \sum_i V(|\psi_i|^2).$$

It is globally regular, positive and exactly invariant under local vertex phases. Literal variation gives one coherent covariant Euler equation, and $U = 1$ recovers the strict A. Thus a minimal conditional gauge-matter repair exists directly from the Dirichlet form.

Immediate falsification shows why this is not yet strict physics. The same weights admit $U(1)$ with every integer charge, Z_N, $SU(2)$, $U(k)$ and real non-gauge models. Complexification, gauge group, representation, connection state and radial-potential coefficients are not selected by A. First-order factorizations are nonunique unless an edge-locality axiom is added.

Gauge dynamics is more obstructed. The K_{12} one-skeleton has 66 edges and cycle rank 55. Without 2-cells it has 55 independent holonomies and no local curvature action. There are 220 triangle cliques; their boundary map has rank 55, and the full clique complex is the contractible 11-simplex. Adding every triangle removes first homology but still leaves arbitrary face areas and Wilson coefficients. Thresholding to the nearest-neighbour C_{12} creates H^1 of dimension one only by discarding 54 nonzero strict edges. The support graph has diameter one, so it does not supply extended locality or a 3+1-dimensional Maxwell limit.

The exact refinement $A_{24} = A_{12} \otimes I_2 + I_{12} \otimes B_q$ transports the normalized fiber-constant matter action, but grows the cycle rank from 55 to 121, adds relative fiber phases, and leaves q, lengths, face areas and scaling laws free. It does not select a continuum.

A first-order fermionic test gives another split result. No nontrivial vertex-diagonal grading anticommutes with a full-support vertex Dirac operator: triangles make the sign equations inconsistent. Doubling vertices and edges produces an even incidence Dirac operator with square containing A, but the 78-dimensional space has 56 zero modes (one constant vertex mode and 55 cycle modes). Its index equals the graph Euler characteristic $12 - 66 = -54$, not a derived particle-generation index. Edge-algebra representation, real structure, first-order condition, fermion content, cutoff scale and gravity remain additional data.

The direct gauge action therefore passes important conditional subrows of the source–action gate but zero complete strict rows. The highest-value next theorem is now sharply identified: derive or refute an automorphism-natural, refinement-functorial weighted 2-complex and face measure from FIN data. Such an object is required before a strict gauge kinetic action, locality or continuum physics can be claimed.

Contents

1 Starting point after Release 11.03

Release 11.03 established that the APD quotient, three moment features and the current mixed-lineage L_{total} do not derive unique physical dynamics. The surviving unavoidable variational object is

$$S_A[f] = \frac{1}{2} f^T A f.$$

The present cycle asks whether the first-order square root of this form can supply a better source action without fitting effective couplings.

The test is intentionally stronger than constructing a gauge-looking formula. Every positive construction is immediately checked for:

1. strict source and uniqueness;
2. a gauge kinetic/curvature term;
3. refinement naturality and continuum scaling;
4. chiral fermionic and gravitational content;
5. compatibility with the seven-row source–action gate.

2 Round I: ST1812–ST1826 — direct gauge-covariant action

Orient every unordered pair once and define

$$(d_W f)_{ij} = \sqrt{w_{ij}}(f_i - f_j), qquad i < j.$$

The edge-sum identity gives

$$d_W^* d_W = A$$

to numerical residual below 10^{-15} in the independent reconstruction. Orientation reversal changes a row convention but not the norm.

For unit link phases U_{ij} with $U_{ji} = \overline{U}_{ij}$ define

$$(d_U \psi)_{ij} = \sqrt{w_{ij}}(\psi_i - U_{ij}\psi_j).$$

Under

$$\psi_i \mapsto g_i \psi_i, qquad U_{ij} \mapsto g_i U_{ij} g_j^{-1}, qquad g_i \in U(1),$$

one has

$$(d_{U'}\psi')_{ij} = g_i (d_U \psi)_{ij}.$$

Therefore

$$S_{\text{matter}} = \frac{1}{2}\|d_U\psi\|^2 + \sum_i V(|\psi_i|^2)$$

is exactly gauge invariant. The covariant Laplacian $A_U = d_U^* d_U$ is Hermitian positive semidefinite, and $U = 1$ gives A.

Literal variation yields

$$A_U \psi + V'(|\psi|^2)\psi = J.$$

A random finite gauge transformation reproduced the action exactly in double precision. The configuration space $U(1)^{66} \times \mathbb{C}^{12}$ contains no APD poles.

Theorem 1 (conditional minimal gauge-matter completion)**.** *After choosing a complex scalar, local $U(1)$ symmetry with unit charge, one link per nonzero edge and radial potential, the weighted covariant difference is a globally regular positive gauge-invariant completion of the strict Dirichlet matter action.*

The theorem does not contain gauge-field kinetic dynamics.

3 Round II: ST1827–ST1841 — uniqueness falsification

The flat connection $U = 1$ erases the group and charge information. The same Dirichlet weights support:

- $U(1)$ with every integer charge via U_{ij}^q;
- discrete Z_N link fields;
- non-Abelian $SU(2)$ or $U(k)$ links in any unitary representation;
- the original real classical/Markov theory without complex phase.

All reduce to the same A at the identity connection.

The factorization itself has edge-Hilbert freedom:

$$A = d_W^* d_W = (Od_W)^*(Od_W)$$

for every edge-space unitary/orthogonal O. Requiring one edge degree of freedom per nonzero pair and support locality reduces this to harmless orientation/phase conventions, but that requirement is an additional calculus axiom.

Because all off-diagonal entries are nonzero, the support graph K_{12} is canonical at the support level. However, A contains no nontrivial link phases, holonomy state, charge, group or scalar representation. Gauge covariance constrains the form of radial potentials but not their coefficients.

Thus the direct action is the minimal repair *conditional* on a chosen gauge principle; it is not a unique physical consequence of FIN.

4 Round III: ST1842–ST1856 — curvature and topology

The complete support has

$$V = 12, qquad E = 66, qquad b_1 = E - V + 1 = 55.$$

An edge connection modulo vertex gauge therefore has 55 independent cycle holonomies if no faces are specified.

There are

$$\binom{12}{3} = 220$$

triangle cliques. Exact integer linear algebra gives rank 55 for the triangle-to-edge boundary matrix. Attaching all triangles kills the one-skeleton H^1; attaching all higher cliques gives the full 11-simplex, which is contractible.

A Wilson family can be written as

$$S_g = \sum_f \kappa_f (1 - \operatorname{Re} U_f),$$

but A does not select the face set, face orientation, area or coefficients κ_f. Products, geometric means, minima and external areas all define different positive face measures.

Thresholding cyclic distance gives another nonunique topology. Keeping only distance one yields C_{12} with 12 edges and $b_1 = 1$, but discards 54 strict nonzero edges. Increasing the retained distance changes the edge and cycle counts at every step. No threshold is selected by nonzero support.

Theorem 2 (support-to-curvature nonuniqueness)**.** *The strict weighted support determines neither a unique 2-complex nor a unique local gauge kinetic action. The full support has graph diameter one and does not by itself define extended locality or a 3+1-dimensional Maxwell limit.*

5 Round IV: ST1857–ST1871 — refinement

For $q > 0$, the exact two-fiber refinement has

$$A_{24} = A_{12} \otimes I_2 + I_{12} \otimes B_q,$$

with 24 vertices, 144 edges and cycle rank

$$144 - 24 + 1 = 121.$$

Its weighted incidence factorization reproduces A_{24} to machine precision.

Copy a base connection to both fibers, set the fiber connection flat, and use the normalized constant fiber vector $u_+ = (1, 1)/\sqrt{2}$. Then

$$(\psi \otimes u_+)^* A_{24} (\psi \otimes u_+) = \psi^* A_{12} \psi.$$

Thus the matter action transports exactly.

The full fine gauge group has 24 local phases rather than 12. Twelve relative fiber phases and twelve fiber link variables are invisible to the coarse factorized subgroup. The rate q remains free. Each further binary refinement introduces at least one new rate and relative connection layer unless a consistency law is supplied.

The refined support is $K_{12} \square K_2$, with degree 12 and diameter two. It remains highly nonlocal rather than approaching an extended continuum. No scaling law for q, edge lengths, face areas, effective dimension or causal speed is exported.

The positive conclusion is exact refinement transport of matter dynamics. The negative conclusion is that refinement does not select locality, topology, dimension or continuum gauge physics.

6 Round V: ST1872–ST1886 — Dirac, chirality and gravity

A naive symmetric vertex operator built from $\sqrt{w_{ij}}$ does not square to A. More importantly, suppose a vertex-diagonal grading $\Gamma = \operatorname{diag}(\gamma_i)$ with $\gamma_i = \pm 1$ anticommutes with a full-support vertex Dirac operator. Every nonzero edge requires $\gamma_j = -\gamma_i$. A triangle then gives a contradiction. Therefore no nontrivial vertex-diagonal even grading exists on the complete support.

A standard first-order repair doubles vertices and edges:

$$\mathcal{D} = \begin{pmatrix} 0 & d_W^* \\ d_W & 0 \end{pmatrix}, qquad\Gamma = \begin{pmatrix} I_{12} & 0 \\ 0 & -I_{66} \end{pmatrix}.$$

Then $\{\Gamma, \mathcal{D}\} = 0$ and

$$\mathcal{D}^2 = operatornamediag(A, d_W d_W^*).$$

The 78-dimensional operator has 22 nonzero modes and 56 zero modes: one constant vertex mode and 55 edge-cycle modes. Its incidence index is

$$1 - 55 = 12 - 66 = -54,$$

the graph Euler characteristic, not a derived physical generation count.

This even complex is not yet a real spectral triple. A vertex-function algebra has no unique action on unoriented edges: source, target, average and doubled-oriented representations differ. Reality, first-order condition and orientation cycle are missing. Inner fluctuations require those choices.

A spectral action further needs a cutoff function, dimensional cutoff, fermionic representation and gauge kinetic normalization. The incidence Dirac does not supply a 3+1 spin structure, Lorentzian signature, metric tensor or diffeomorphism/gravity action.

7 Round VI: ST1887–ST1901 — final gate and verdict

Object	Status	Boundary
$A = d_W^* d_W$	[Proven]	finite weighted graph
local $U(1)$ matter action	[Proven conditional]	chosen complex field/group/links
gauge group, charge and connection	[Not sourced]	many exact alternatives
canonical gauge kinetic action	[Refuted from support alone]	face/area ambiguity
coarse refinement transport	[Proven]	factorized connection/constant fiber
continuum dimension/locality	[Not derived]	free scaling/moduli
even incidence Dirac	[Constructed]	56 zero modes
real/chiral SM spectral triple	[Not derived]	algebra/reality/representation missing
gravity/physical gauge theory	[Not established]	scale, topology, platform absent

The updated strict gauge-source gate contains seven rows:

H1. strict state-category/gauge-group/charge source;

H2. strict connection/holonomy source;

H3. canonical 2-complex, face measure and gauge kinetic action;

H4. refinement scaling to local continuum dimension and causality;

H5. real even spectral triple with chiral fermion representation;

H6. dimensional action and nonabsorbable prediction;

H7. OA platform and independent record.

The direct action makes conditional progress on regularity, gauge covariance and literal variation, but zero rows are complete in the strict sense.

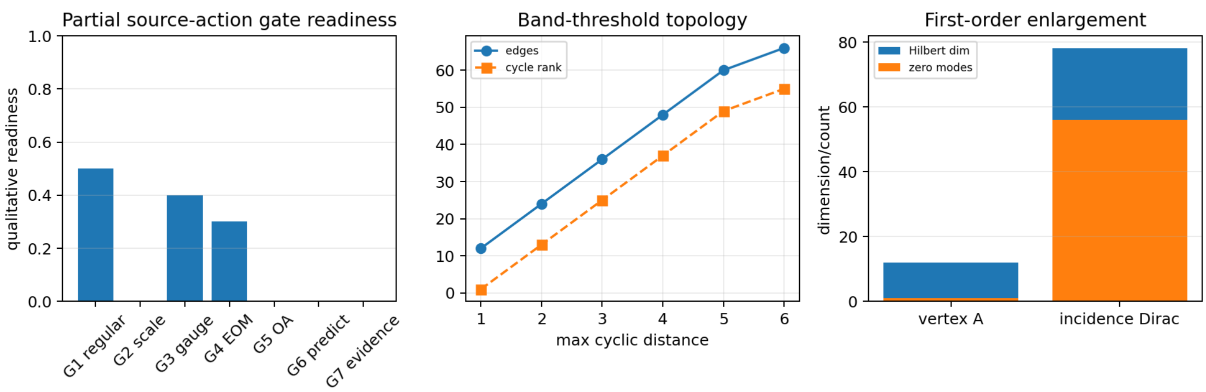

Figure 1: Left: qualitative partial readiness of the prior source–action gate. Centre: topology changes under distance-band thresholding. Right: the first-order incidence construction enlarges the state space and creates 56 zero modes.

8 Most important result

The most important result is the constructive/negative pair:

$$\boxed{\text{strict Dirichlet data admits a clean conditional gauge-matter action,}}$$

but

$$\boxed{\text{it does not select gauge group, curvature complex, continuum or chiral physics.}}$$

This is more informative than the earlier L_{total} scaffold. Gauge invariance and one-action EOM can be repaired without APD or moment fitting, but the repair exposes exactly which new objects are still absent.

9 Highest-value next theorem

The single most decisive next target is:

> *derive or refute an automorphism-natural, refinement-functorial map from strict FIN data to an oriented weighted 2-complex with positive face measure, whose Wilson or spectral gauge action has a controlled local continuum limit.*

It should be falsified in this order:

1. automorphism naturality;
2. independence of arbitrary thresholds;
3. refinement functoriality;
4. local dimension and causal scaling;
5. positive face weights;
6. continuum Maxwell normalization.

If no such functor exists from the full strict/refinement data, the gauge-field route from FIN is blocked without new physics by assumption. If it does exist, it would close the most important gap left by this cycle and provide a real input for a spectral/source action.

10 Route ranking

Route	Rigorous-success estimate	Physics decision power
derive canonical 2-complex/face measure from refinement	0.28	1.00
publish conditional $U(1)$ Dirichlet lattice matter model	0.80	0.45
complete real even spectral triple from incidence complex	0.30	0.75
continue APD/moment L_{total} repair	0.01	0.50
OA laboratory test of frozen channels	0.45	0.55

These estimates are strategic judgments, not empirical probabilities.

11 Final interpretation

The direct first-order Dirichlet construction is currently the most promising mathematical bridge from FIN toward gauge theory because it is globally regular, positive, exactly gauge covariant and variationally coherent. It is still conditional rather than strict: the group, charge, connection state, 2-complex, face weights, scale, fermions and platform are not derived. FIN therefore remains a finite Dirichlet–Markov–spectral information geometry with a viable conditional lattice gauge-matter extension, not a fundamental physical theory.

A Six-round ledger

Range	Target	Verdict
ST1812–ST1826	covariant Dirichlet matter action	exact conditional construction
ST1827–ST1841	uniqueness	group/charge/calculus nonuniqueness
ST1842–ST1856	curvature topology	no canonical 2-complex/Maxwell action
ST1857–ST1871	refinement	matter transports; continuum not selected
ST1872–ST1886	Dirac/fermions/gravity	even complex, large zero sector, physics open
ST1887–ST1901	synthesis	updated seven-row gauge-source gate

B Reproducibility

The authoritative result sets run from `FIN_ST1812_ST1826_Results.json` through `FIN_ST1887_ST1901_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1812_st1901.py` verifies all ninety packets, gauge invariance, topological ranks, refinement factorization, grading/zero modes, updated gate, figure and nonpromotion boundaries.

C Nonclaims

This report does not derive the gauge group, charge, connection state, Wilson face action, continuum spacetime, chirality, fermion generations, gravity, physical units, state ontology, platform or evidence. It does not complete legacy role transfer, QW-2191, Standard Model, L_{total} or Theory of Everything. All future FIN PDFs remain English.

References

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